Section B

6 (a) (i) The linear momentum *p* of a body is the product of its mass *m* and instantaneous linear velocity *v*.

(ii) The rate of change of momentum of a body is proportional to the net force that acts on it and has the same direction as the net force.

$$F_{net} \propto \frac{dp}{dt}$$

Note: 'rate of change of momentum *per unit time*' is not correct.

(b) (i) The total momentum of a system is constant, provided no external resultant force acts on it.

Note: provided 'no external force' is incorrect as resultant force is crucial.

(ii) It is not possible for both trolleys to be at rest simultaneously as this will mean that the total momentum at that instance is zero. This will violate the principle of conservation of momentum as the total momentum before collision is not zero, since both trolleys are moving in the same direction initially.

(c) (i) Kinetic energy $= \frac{1}{2}mv^2 = \frac{p^2}{2m} = 0.74$ MeV

$$p = \sqrt{2(9.11\times10^{-31})(0.74\times10^{6})(1.6\times10^{-19})}$$

$$= 4.64\times10^{-22} \text{ kg m s}^{-1}$$

(ii) 1. $E = hf = \frac{hc}{\lambda}$

$$\lambda = \frac{hc}{E} = \frac{(6.63\times10^{-34})(3.0\times10^{8})}{(0.85\times10^{6})(1.6\times10^{-19})}$$

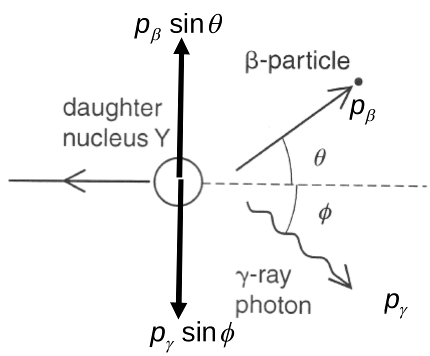
$$= 1.46\times10^{-12} \text{ m}$$

2. $p = \frac{h}{\lambda}$

$$= \frac{6.63\times10^{-34}}{1.46\times10^{-12}} = 4.53\times10^{-22} \text{ kg m s}^{-1}$$

(d) (i) Initial momentum of X is zero as it is initially stationary and since the total momentum of the system is constant as no external resultant force acts on it, the total momentum after decay is also zero.
Thus, if the daughter nucleus Y is stationary, then the only way for total momentum of the system to be zero is that β - particle and γ - particle move off in opposite direction so that their momentum is opposite. As calculated in **(c)**, as their momentums are not exactly equal, they will move off in approximately opposite direction.

(ii) Since total momentum is zero, the vertical components of the momentums of the β - particle and γ - particle must be equal to each other as shown below.



Resolving the momentum vertically,

$$p_\beta \sin\theta = p_\gamma \sin\phi \Rightarrow \sin\theta = \frac{p_\gamma}{p_\beta}\sin\phi$$